

12.4

Disk Scheduling

(a) FCFS

- The simplest form of disk scheduling.
- Generally does not provide the fastest service.

Example 3 :

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We illustrate the **FCFS** disk scheduling algorithms with the following information to get the total distance (in cylinders) :

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.



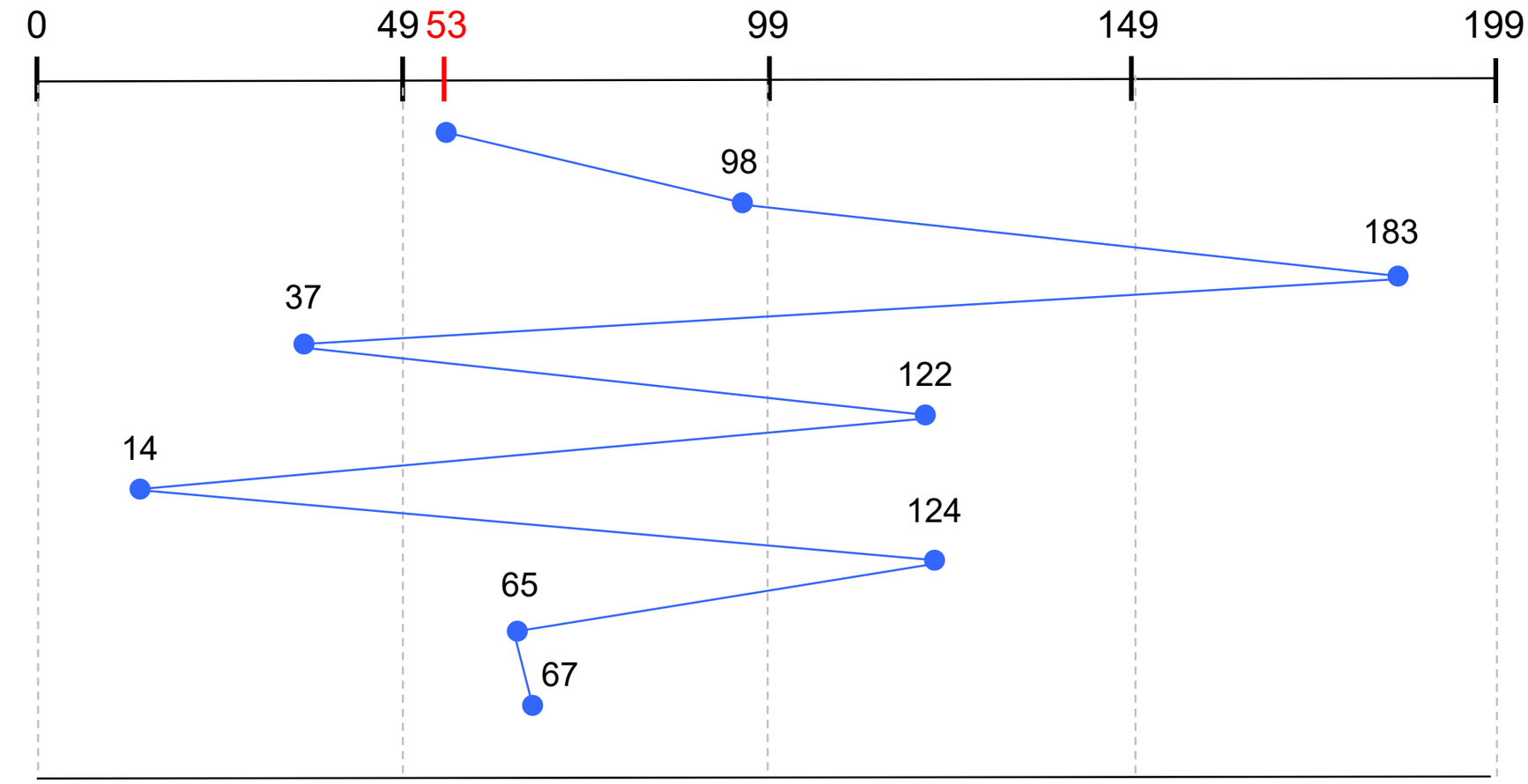
Disk Scheduling

Solution 3 :

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- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(98-53)+$

$(183-98)+$

$(183-37)+$

$(122-37)+$

$(122-14)+$

$(124-14)+$

$(124-65)+$

$(67-65)$

$=45+85+146+85+108+110+59+2=640$ cylinders.



Disk Scheduling

Shortest Seek Time First (SSTF)

(b) SSTF

- This algorithm selects the request with the minimum seek time from the current head position.
- SSTF scheduling is a form of SJF scheduling; may cause starvation of some requests.
- Next example illustrates a total head movement of 236 cylinders.





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Disk Scheduling

Example 4 :

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We illustrate the **SSTF** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



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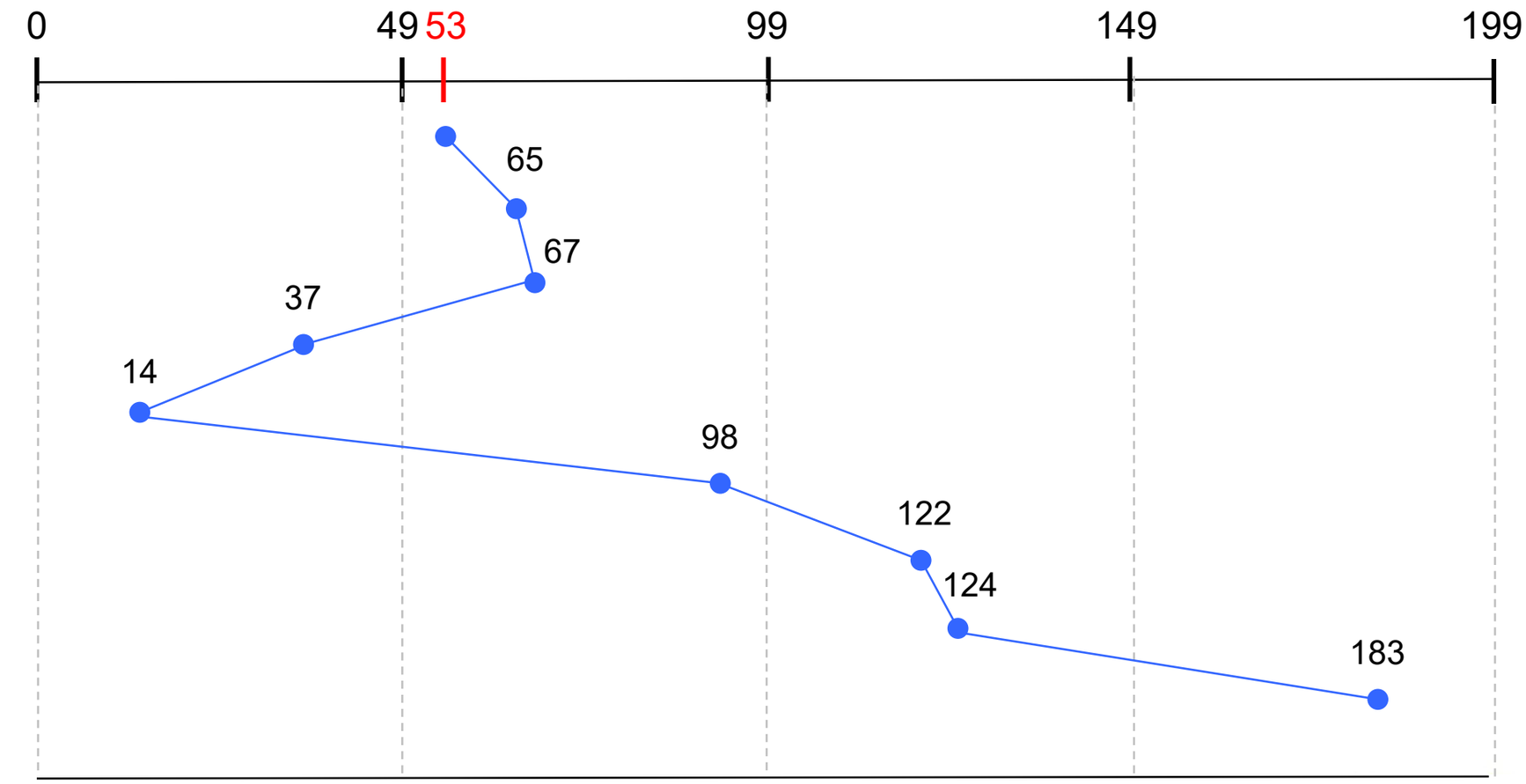
Disk Scheduling

Solution 4 :

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- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65-53)+$

$(67-65)+$

$(67-37)+$

$(37-14)+$

$(98-14)+$

$(122-98)+$

$(124-122)+$

$(183-124)$

$=12+2+30+23+84+24+2+59=236$ cylinders (head movement).



Disk Scheduling

- Although **SSTF** algorithm is substantial improvement over **FCFS** algorithm, it is not optimal.
- We can do better by moving the head from 53 to 37, even though the latter is not closest.
- This strategy reduces the total head movement to 208 cylinders.

Exercise 12.1 : Based on previous example, redraw the entire head movement of the disk if the head moving from 53 to 37 instead of 53 to 65.



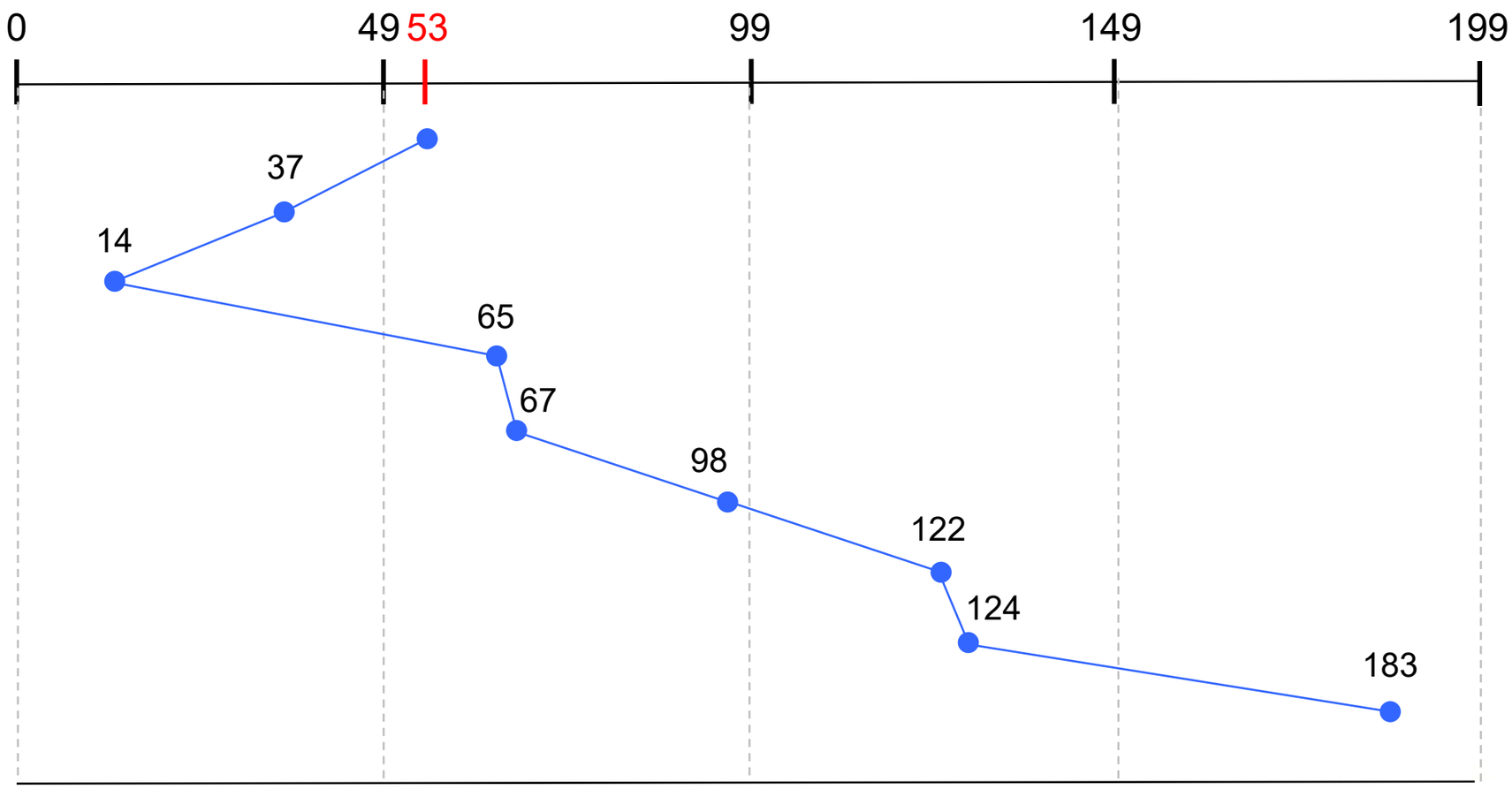
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Disk Scheduling

Solution 12.1

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$$(53-37)+$$
$$(37-14)+$$
$$(65-14)+$$
$$(67-65)+$$
$$(98-67)+$$
$$(122-98)+$$
$$(124-122)+$$
$$(183-124)$$

$$=16+23+51+2+31+24+2+59=208$$ cylinders (head movement).



Disk Scheduling

(c - i) SCAN

- The disk arm starts at one end of the disk, and moves **toward** the other end, servicing requests until it gets to the other end of the disk, where the head movement is **reversed** and servicing continues.
- **SCAN algorithm** sometimes called the **elevator algorithm.**

- Next example illustrates total head movement of 208 cylinders.
- But note that if requests are uniformly dense, largest density at other end of disk and those wait the longest.





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Disk Scheduling

Example 5 :

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We illustrate the **SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving toward 0.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



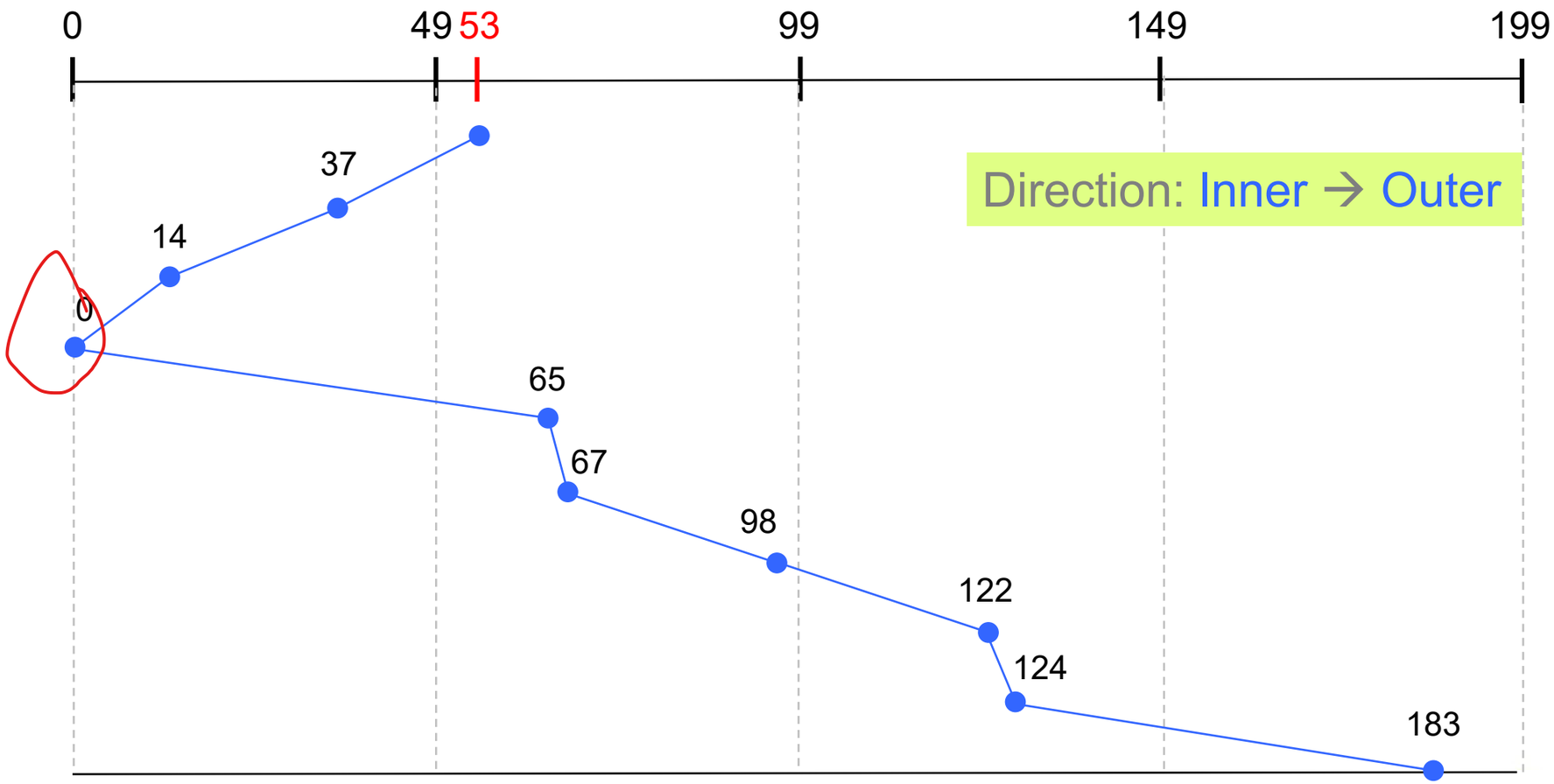
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Disk Scheduling

Solution 5 :

- ✓ A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(53-37)+$

$(37-14)+$

$(14-0)+$

$(65-0)+$

$(67-65)+$

$(98-67)+$

$(122-98)+$

$(124-122)+$

$(183-124)$

$=25+23+14+65+2+31+24+2+59=245$ cylinders (head movement).



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Disk Scheduling

Exercise 12.2 :

We illustrate the **SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving to innermost.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



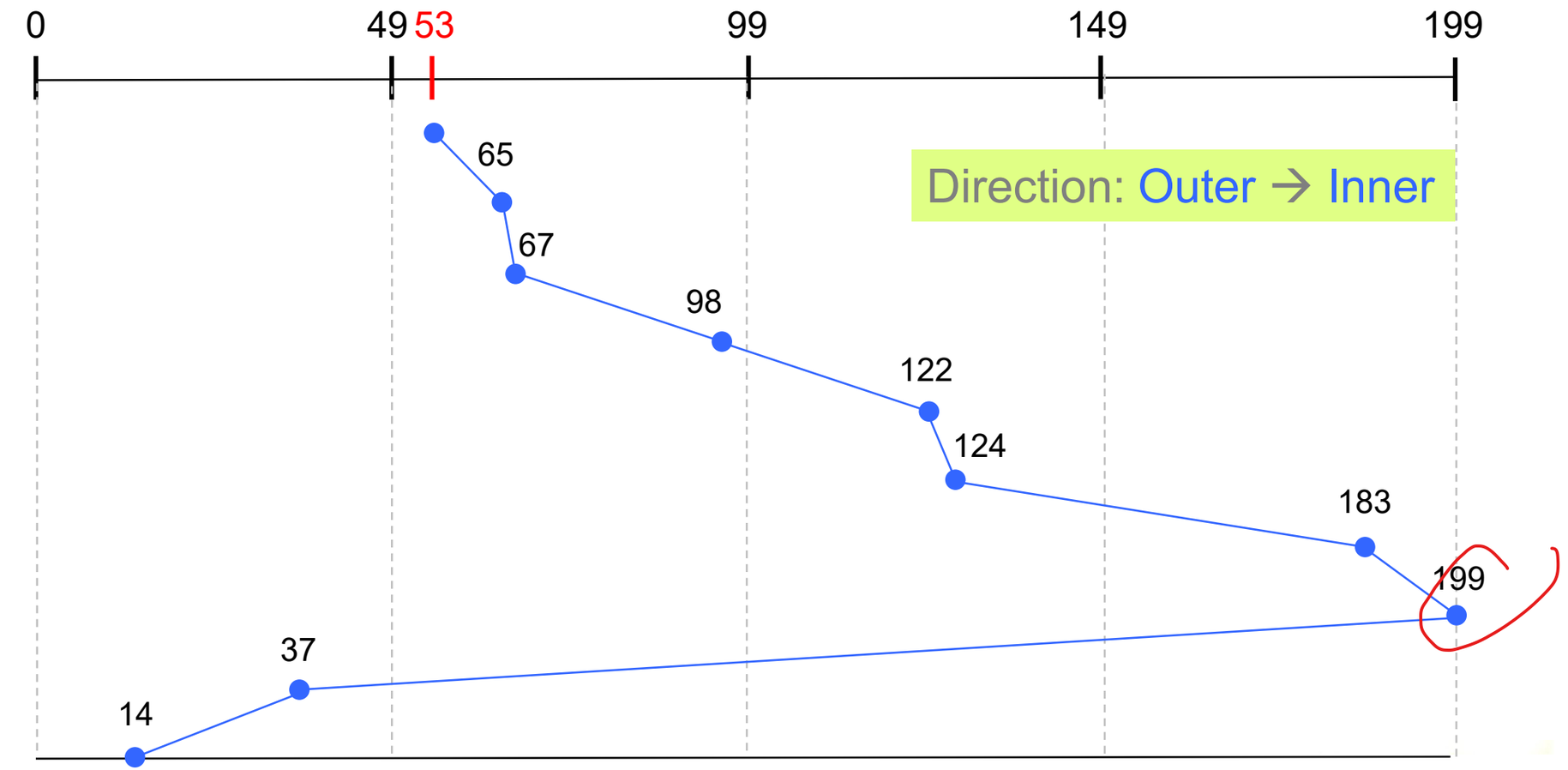
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Disk Scheduling

Solution 12.2 :

- ✓ A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65-53)+$

$(67-65)+$

$(98-67)+$

$(122-98)+$

$(124-122)+$

$(183-124)+$

$(199-183)+$

$(199-37)+$

$(37-14)$

$=12+2+30+23+84+24+2+59=331$ cylinders (head movement).



Disk Scheduling

(c-ii) C-SCAN

- Provides a more uniform wait time than **SCAN**.
- The head moves from one end (**inward sweep**) of the disk to the other, servicing requests as it goes:
  - When it reaches the other end, however, it immediately **returns** to the beginning of the disk, without servicing any requests on the return trip.
- Treats the cylinders as a circular list that wraps around from the last cylinder to the first one.



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Disk Scheduling

Example 6 :

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We illustrate the **C-SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



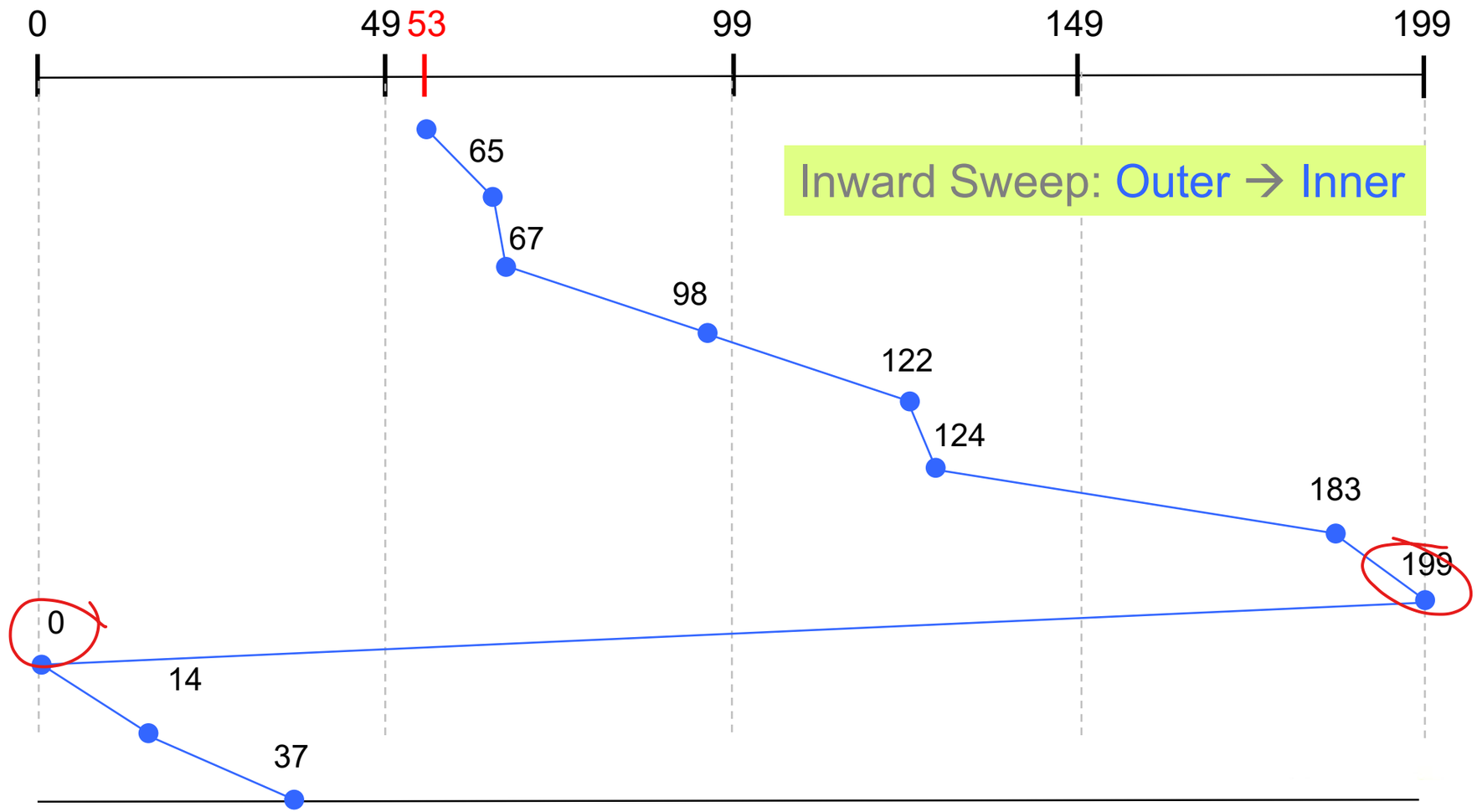
Disk Scheduling

Solution 6 :

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- ✓ A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$$(65-53)+$$
$$(67-65)+$$
$$(98-67)+$$
$$(122-98)+$$
$$(124-122)+$$
$$(183-124)+$$
$$(199-183)+$$
$$(199-0)+$$
$$(14-0)+$$
$$(37-14)$$

$$=12+2+31+24+2+59+16+199+14+23=382$$ cylinders (head movement).



Disk Scheduling

(d) LOOK and C-LOOK

- As described before, both SCAN and C-SCAN move the disk arm across the full width of the disk.
- The arms goes only as far as the final request in each direction, then reverses direction immediately, without going all the way to the end of the disk.
- Version of SCAN and C-SCAN that follow this pattern is called **LOOK** and **C-LOOK** disk scheduling.





Disk Scheduling

Example 7 :

We illustrate the **LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving toward 0.

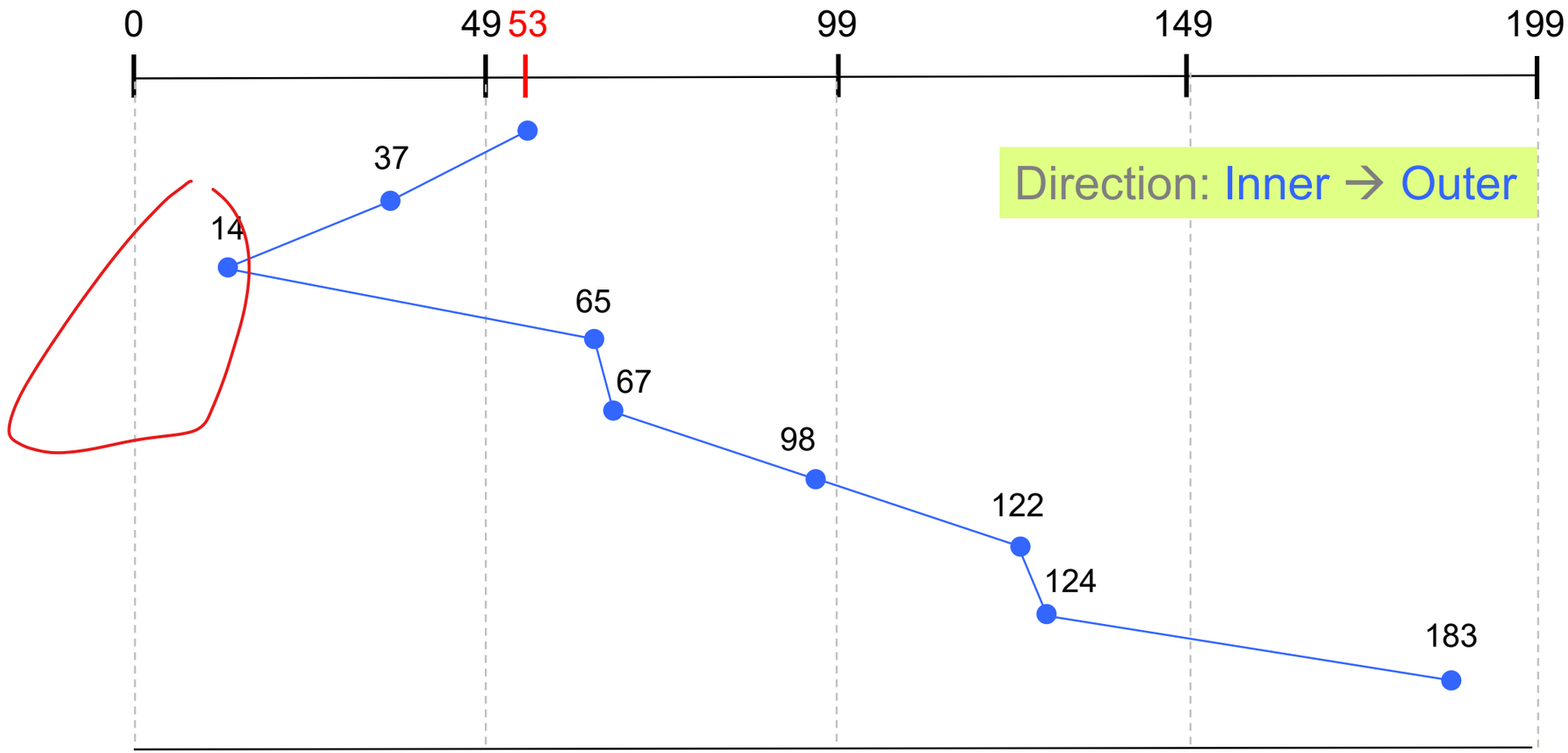
Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.

Disk Scheduling

Solution 7 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$$(53-37)+$$
$$(37-14)+$$
$$(65-14)+$$
$$(67-65)+$$
$$(98-67)+$$
$$(122-98)+$$
$$(124-122)+$$
$$(183-124)$$

$$=16+23+51+2+31+24+2+59=208$$ cylinders (head movement).



Disk Scheduling

Exercise 12.3 : We illustrate the **LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving to innermost.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



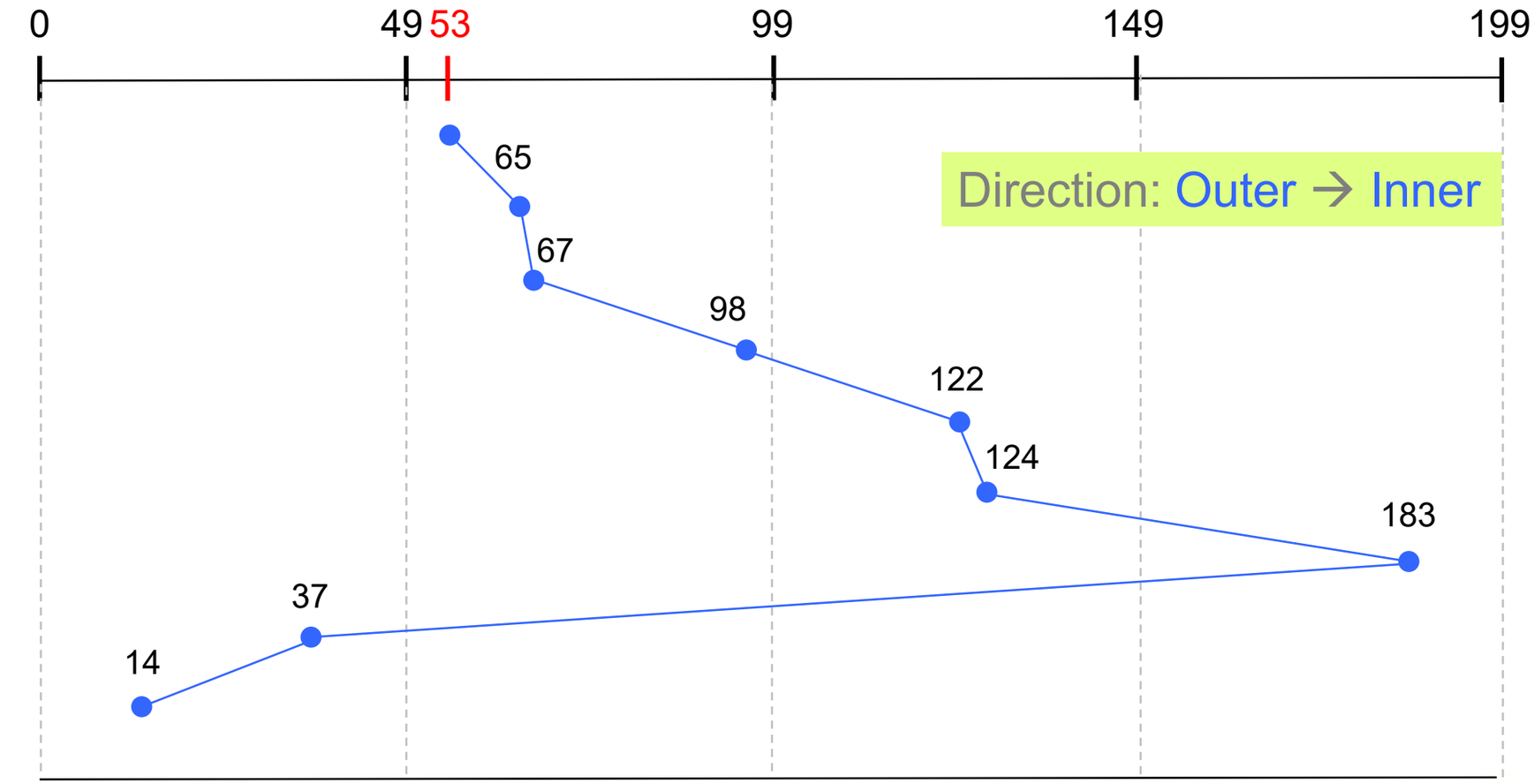
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Disk Scheduling

Solution 12.3 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$$(65-53)+$$
$$(67-65)+$$
$$(98-67)+$$
$$(122-98)+$$
$$(124-122)+$$
$$(183-124)+$$
$$(183-37)+$$
$$(37-14)$$

$=12+2+30+23+84+24+2+59=331$ cylinders (head movement).



Disk Scheduling

Example 8 :

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We illustrate the **C-LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



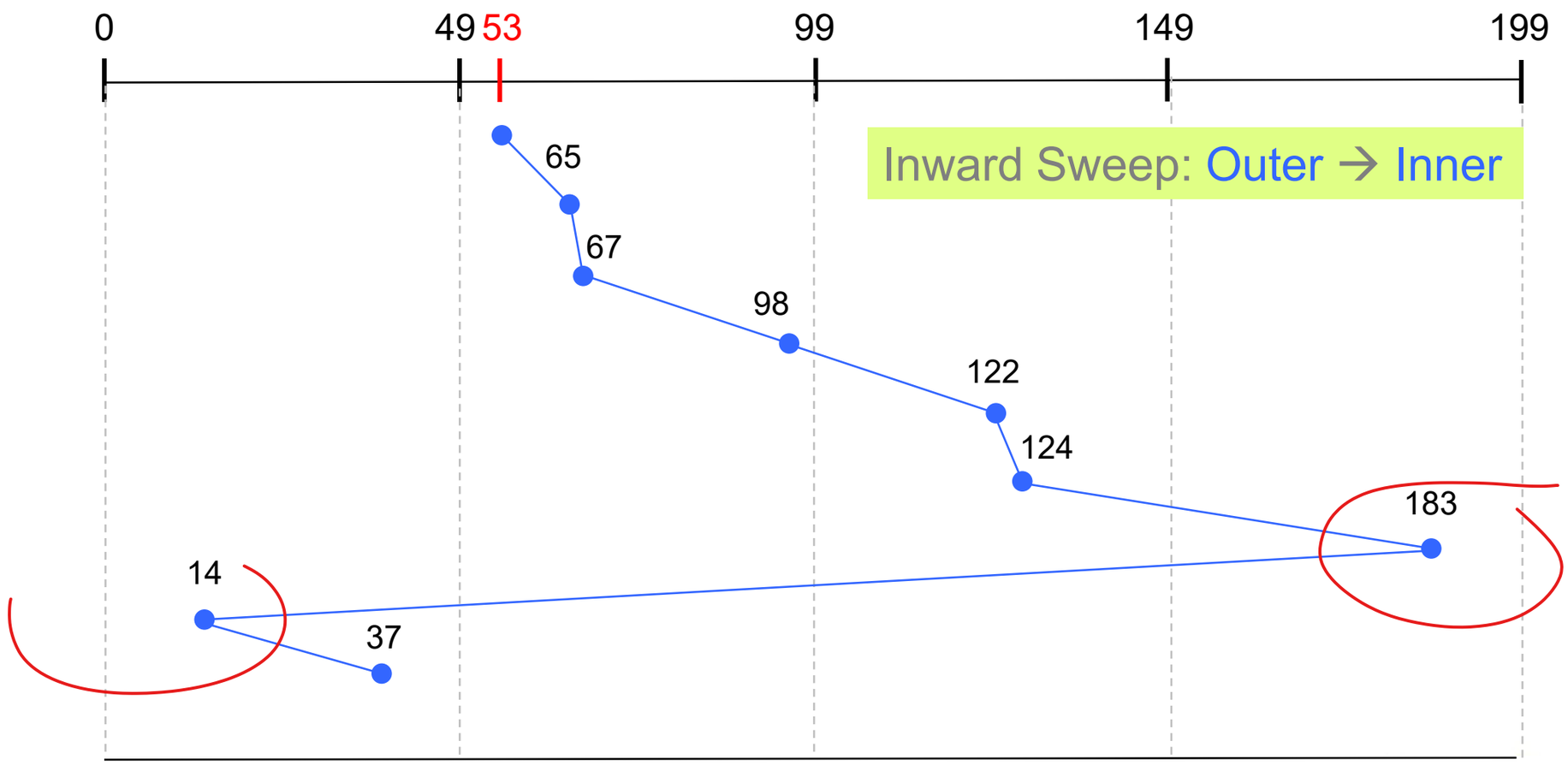
Disk Scheduling

Solution 8 :

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- ✓ A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$$(65-53)+$$
$$(67-65)+$$
$$(98-67)+$$
$$(122-98)+$$
$$(124-122)+$$
$$(183-124)+$$
$$(183-14)+$$
$$(37-14)$$

$$=12+2+31+24+2+59+169+23=322$$ cylinders (head movement).